

Metro Ridership Forecasting using Inter-Station-Aware Transformer Networks

Big Data Course Presentation (10 minutes max)

Nima DAVARI Maksym DOLHOV Khanh Bao ba Mehdi AGHAEI

Big Data Course



[1/12] Problem Context (Presenter 1)

- ▶ Metro systems need station-level inflow/outflow forecasts for operations.
- ▶ Better forecasting improves train scheduling, staffing, maintenance, and capacity planning.
- ▶ Wrong forecasts can increase congestion and reduce service quality.
- ▶ The task is hard because demand is both **temporal** and **network-dependent**.



[2/12] Research Gap and Paper Idea (Presenter 1)

- ▶ Prior methods include statistical, machine learning, and hybrid approaches.
- ▶ Many RNN/GCN-based methods are strong for short horizon but can degrade for longer horizon.
- ▶ This paper proposes **ISA-TF**: a unified transformer framework for metro ridership forecasting.
- ▶ Core idea: fuse ridership history with inter-station topology information.



[3/12] Claimed Contributions (Presenter 1)

- ▶ New **Inter-Station-Aware Transformer Networks** architecture.
- ▶ Uses three metro graph views: physical, semantic similarity, and correlation.
- ▶ Evaluated on SHMetro and HZMetro benchmarks.
- ▶ Reports stronger long-horizon performance and much lower parameter count.



[4/12] Problem Formulation (Presenter 2)

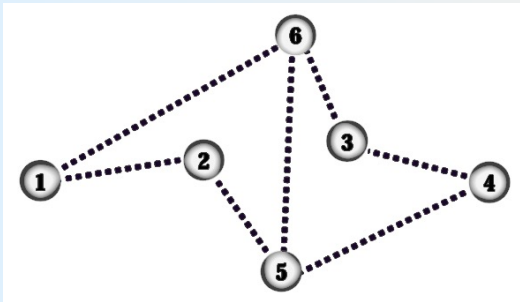
- ▶ Input: past ridership sequence and metro graph:

$$\{D_{t-n+1}^N, \dots, D_t^N\}, G \rightarrow \{\hat{D}_{t+1}^N, \dots, \hat{D}_{t+m}^N\}$$

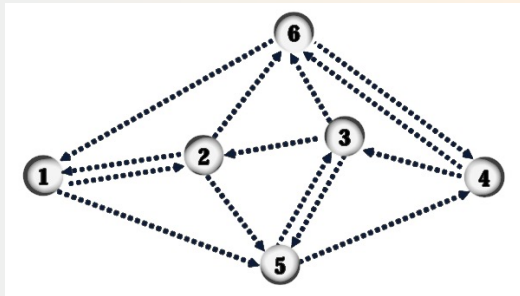
- ▶ Each station signal contains two values: inflow and outflow.
- ▶ In experiments: $n = 4$ past intervals (60 min total), $m = 4$ future intervals.
- ▶ Inference is autoregressive, so horizons longer than 60 min are possible.



[5/12] Metro Graph Representation (Presenter 2)



Physical topology view



Enhanced inter-station relation view

Explanation: The paper represents the metro network with complementary graph structures, not only direct neighboring links. This helps the model capture both local structure and richer cross-station interaction patterns, which improves forecasting robustness when passenger flow dependencies are not purely geographic.



- ▶ Encoder-decoder transformer backbone with positional encoding.
- ▶ Encoder processes historical ridership tokens.
- ▶ Intermediate fusion concatenates encoder output with M_p , M_s , M_c , then feed-forward layer.
- ▶ Decoder uses self-attention and cross-attention to generate future ridership autoregressively.



Property	SHMetro	HZMetro
Period	Jul–Sep 2016	Jan 1–25, 2019
Stations	288	80
Physical edges	958	248
Avg daily passengers	8.82M	2.35M
Interval	15 minutes	15 minutes

SHMetro has 811.8M recorded transactions.



[8/12] Experimental Setup (Presenter 3)

- ▶ Preprocessing creates the three graph types per dataset.
- ▶ Training details: Adam optimizer, 250 epochs, L_2 loss.
- ▶ Model hyperparameters: embedding size 512, 8 attention heads.
- ▶ Evaluation metrics: RMSE and MAE.
- ▶ Baselines: HM, GB, MLP, RNN-GRU, STG2Seq, DCRNN, PVCGRN.



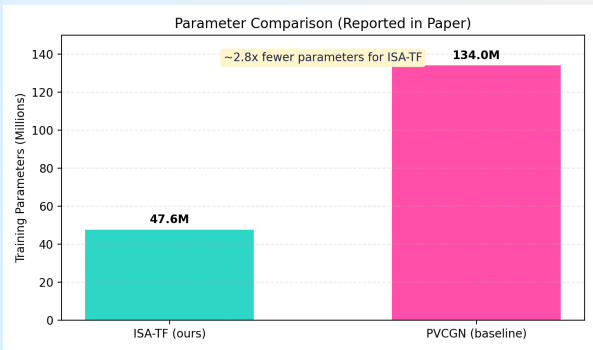
[9/12] Main Results (Presenter 3)

Dataset	Horizon	ISA-TF (RMSE/MAE)	PVCGN (RMSE/MAE)
SHMetro	60 min	48.43/23.87	55.27/26.29
HZMetro	60 min	41.68/24.91	42.61/24.93
SHMetro	15 min	45.85/ 23.05	44.97 /23.29

- ▶ Strong long-horizon robustness is the key result.
- ▶ Short horizon remains competitive with the prior SOTA.



[10/12] Efficiency Comparison (Presenter 4)



Explanation: Based on the paper's reported values, ISA-TF uses 47.6M parameters while PVCGN requires about 134M. This means ISA-TF is around 2.8x lighter, which is important for faster training, lower memory use, and easier deployment while keeping competitive or better forecasting performance.

Source: values reported by the paper (Figure 2 discussion and results section).



[11/12] Discussion for Class (Presenter 4)

- ▶ **Strengths:** unified architecture, good long-horizon behavior, parameter efficiency.
- ▶ **Caveat:** evaluation is on two datasets only; generalization should be tested further.
- ▶ **Possible extensions:**
 - Include weather/events/holidays as exogenous features.
 - Add uncertainty intervals, not only point forecasts.
 - Evaluate transfer across cities with different network shapes.



- ▶ ISA-TF combines ridership history with topology-aware graph information.
- ▶ It matches or beats prior methods, especially for 45–60 minute horizons.
- ▶ It achieves this with about 3x fewer parameters than PVCGRN.

Takeaway: topology-aware transformers are a strong option for scalable metro ridership forecasting.

Reference: K. Saleh, A.-S. Mihaita, and Y. Ou, 2023 IEEE ITSC (Manuscript 470 preprint).